

Project Title

AI Resume Analyzer Platform

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Objectives

The main objective of this project is to design and develop an AI-based system that can analyze a user's resume and compare it with a given job description. The system will help users understand how suitable their CV is for a specific job and how it can be improved.

The project has the following specific objectives:

- To build a system that can check how well a resume matches a job description using Artificial Intelligence.
- To generate an ATS (Applicant Tracking System) score out of 100 for each resume. This score will show how strong or weak a CV is for a specific job.
- To provide clear and easy feedback that helps users understand what changes are needed in their resume.
- To divide feedback into different parts such as skills, content quality, structure, and writing style so that users can understand each area separately.
- To identify missing or required skills based on the job description.
- To develop a simple and user-friendly web application where resumes can be uploaded and results can be seen easily.
- To help students and job seekers improve their chances of getting selected for jobs and internships by improving their resumes.

Motivation

In today's job market, most companies use ATS (Applicant Tracking System) software to automatically filter CVs before a human recruiter sees them. Because of this system, many qualified candidates are rejected even if they are actually suitable for the job.

This project is chosen for several important reasons:

- Many students in Bangladesh do not clearly understand how ATS systems work in real job applications.

- A large number of students make small mistakes in their CV, such as poor formatting, missing keywords, or weak structure. These mistakes reduce their chances of getting selected.
- There is a need for a system that can guide users step by step to improve their resumes in a simple way.
- Artificial Intelligence can be used to solve real-life problems related to job searching and career development.
- This system is very useful for students who are preparing for internships, campus jobs, or their first job after graduation.
- This project also helps in learning how AI can be applied in practical systems that support real users.

Benefits of the Proposed System

The proposed system provides many useful benefits for users:

- It helps users clearly understand how their resume performs when compared with real job requirements.
- It increases the chance of getting shortlisted for interviews by improving resume quality.
- It provides simple and clear suggestions that help users improve their CV step by step.
- It reduces the need to ask others for manual CV checking or feedback.
- It identifies missing skills and important keywords required for a job position.
- It helps users improve different parts of their resume such as structure, content, and writing style separately.
- It allows users to improve their resume gradually by analyzing multiple versions.
- It is especially helpful for students and fresh graduates in Bangladesh who are preparing for jobs and internships.
- It helps users track their progress by saving previous resume analysis results.

Key Features of the System

The system includes several important features that make it useful and easy to use.

1. AI-Based Resume Analysis

The system uses Artificial Intelligence to analyze the resume content. It compares the resume with a given job description and checks how relevant the CV is for that job. After analysis, it generates an ATS score from 0 to 100. This score helps users quickly understand the strength of their resume.

2. Feedback System

The system provides detailed but easy-to-understand feedback. The feedback is divided into different categories:

- Skills: shows whether required skills are present or missing
- Content quality: checks clarity and relevance of information
- Structure: checks how well the CV is organized
- Writing style: checks professionalism and tone

The system also highlights missing skills based on the job description and gives suggestions to improve the resume.

3. User System

A secure login system is included so that users can create accounts and access their data safely. Each user has a personal dashboard where previous resume analyses are stored. This helps users review their past performance and track improvement over time.

4. File Upload System

Users can upload their resumes in PDF format easily. The system automatically reads and processes the uploaded file. Before analysis, it checks the file format and size to ensure valid input. This makes the system simple and user-friendly.

5. User Interface

The system has a clean and simple interface so that anyone can use it without technical knowledge. It works smoothly on mobile phones, tablets, and desktop computers. The results, scores, and feedback are displayed in a clear and organized way so users can easily understand them.

6. System and Data Handling

The system stores resumes, job details, and analysis results in a cloud-based database. This ensures that user data is safe and can be accessed anytime. The system is also designed in a scalable way so that it can handle multiple users at the same time without performance issues.

Software Process Model

The Agile Software Development Model is selected for this project.

This model is suitable because it allows development to happen step by step instead of building everything at once. Each part of the system can be tested and improved during development.

The reasons for choosing Agile are:

- Development is done in small steps, which makes the project easier to manage.
- Each feature can be tested and improved during development.
- Tasks can be divided among group members easily.
- Changes and improvements can be made anytime during the project.
- It helps in continuous improvement of the system.
- It allows delivery of working parts of the system step by step instead of waiting for the full project to finish.

This project aims to build a useful AI-based resume analysis system that helps users improve their CVs in a simple and effective way. It will increase the chances of getting selected for jobs and internships by providing proper feedback, ATS scoring, and skill gap analysis. The system is designed to be simple, practical, and helpful for students and job seekers.